DSA1604 - Knowledge Engineering and Data Science

Lab Experiments

Experiment 1: Knowledge Representation Using Python

Aim: To represent facts about a student using a Python dictionary and display the stored knowledge.

Software Required: Python 3.x

Procedure / Algorithm

1. Create a dictionary containing student attributes.
2. Traverse the dictionary using items().
3. Display each attribute and its value.

Program

```
student = {
    "name": "Arun",
    "department": "CSE",
    "course": "Python Programming",
    "year": 2
}

print("Student Knowledge Representation")
for key, value in student.items():
    print(key, ":", value)
```

Sample Input

```
Predefined facts: name=Arun, department=CSE, course=Python Programming, year=2
```

Expected Output

```
Student Knowledge Representation
name : Arun
department : CSE
course : Python Programming
year : 2
```

Result: Knowledge was represented as structured facts and displayed successfully.

Student Observation / Remarks: _____

Faculty Signature: _____ Date: _____

Experiment 2: Knowledge Acquisition

Aim: To acquire knowledge from a user and store it as structured facts.

Software Required: Python 3.x

Procedure / Algorithm

4. Read the expert's name, domain, and years of experience.
5. Store the values in a dictionary.
6. Display the acquired knowledge.

Program

```
name = input("Enter expert name: ")
domain = input("Enter area of expertise: ")
experience = int(input("Enter years of experience: "))

knowledge = {
    "expert": name,
    "domain": domain,
    "experience": experience
}

print("\nAcquired Knowledge")
print("Expert:", knowledge["expert"])
print("Domain:", knowledge["domain"])
print("Experience:", knowledge["experience"], "years")
```

Sample Input

```
Enter expert name: Ravi
Enter area of expertise: Networking
Enter years of experience: 8
```

Expected Output

```
Acquired Knowledge
Expert: Ravi
Domain: Networking
Experience: 8 years
```

Result: Knowledge was collected from the user and stored in a structured form.

Student Observation / Remarks: _____

Faculty Signature: _____ Date: _____

Experiment 3: Knowledge-Based Course Recommendation System

Aim: To recommend a course using a small knowledge base and decision rules.

Software Required: Python 3.x

Procedure / Algorithm

7. Read the student's area of interest.
8. Compare the interest with predefined conditions.
9. Display the recommended course or a fallback message.

Program

```
interest = input("Enter your interest (AI, Data, Web): ").strip().lower()

if interest == "ai":
    course = "Artificial Intelligence"
elif interest == "data":
    course = "Data Science"
elif interest == "web":
    course = "Web Development"
else:
    course = "No matching course found"

print("Recommended Course:", course)
```

Sample Input

```
Enter your interest (AI, Data, Web): AI
```

Expected Output

```
Recommended Course: Artificial Intelligence
```

Result: The system used stored knowledge and rules to produce a recommendation.

Student Observation / Remarks: _____

Faculty Signature: _____ Date: _____